
```
clc;
clear;
close all;

% Battery parameters
initial_SOC = 100;      % %
capacity_Ah = 2;        % Ah

% Different discharge currents
current1 = 0.5;          % A
current2 = 1.0;          % A
current3 = 1.5;          % A

% Simulation time
time = 0:0.01:4;

% SOC calculation
SOC1 = initial_SOC - (current1 .* time ./ capacity_Ah) .* 100;
SOC2 = initial_SOC - (current2 .* time ./ capacity_Ah) .* 100;
SOC3 = initial_SOC - (current3 .* time ./ capacity_Ah) .* 100;

% Limit SOC
SOC1(SOC1 < 0) = 0;
SOC2(SOC2 < 0) = 0;
SOC3(SOC3 < 0) = 0;

% Plot
figure;

plot(time, SOC1, 'LineWidth', 2);
hold on;
plot(time, SOC2, 'LineWidth', 2);
plot(time, SOC3, 'LineWidth', 2);

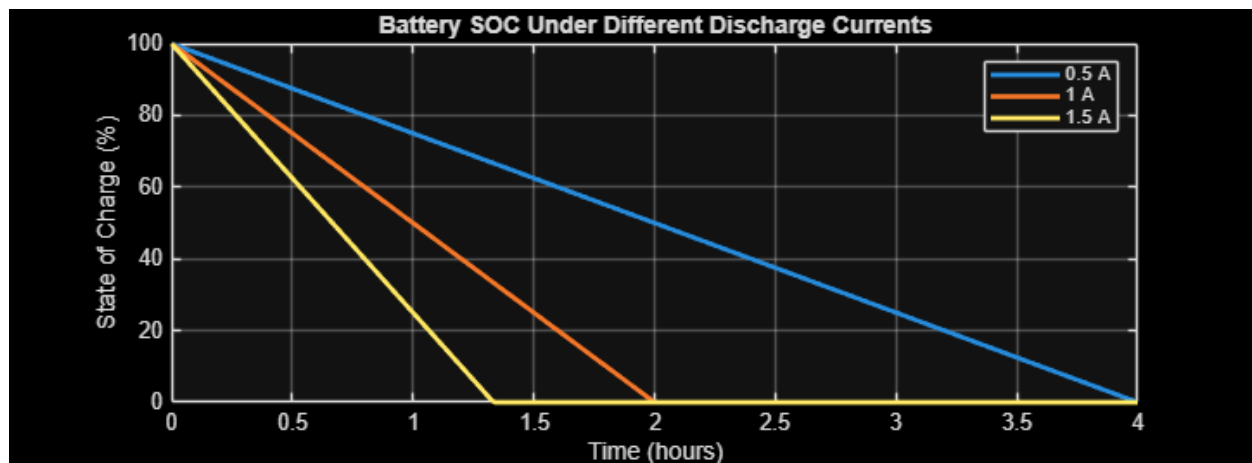
grid on;

xlabel('Time (hours)');
ylabel('State of Charge (%)');

title('Battery SOC Under Different Discharge Currents');

legend('0.5 A', '1 A', '1.5 A');

ylim([0 100]);
xlim([0 4]);
```



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